8 It is given that $y = \cosh u$, where $u > 0$, and

$$\sqrt{\cosh^2 u - 1}\left(\frac{\mathrm{d}^2 u}{\mathrm{d}x^2} + \frac{\mathrm{d}u}{\mathrm{d}x}\right) + \cosh u\left(\frac{\mathrm{d}u}{\mathrm{d}x}\right)^2 - 2\cosh u = 4\mathrm{e}^{-x}.$$

(a) Show that

$$\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} + \frac{\mathrm{d}y}{\mathrm{d}x} - 2y = 4\mathrm{e}^{-x}. \qquad [4]$$

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(b) Find u in terms of x, given that, when $x = 0$, $u = \ln 3$ and $\frac{\mathrm{d}u}{\mathrm{d}x} = 3$. [10]

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